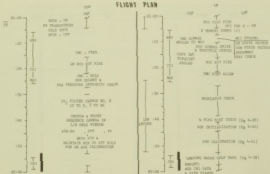
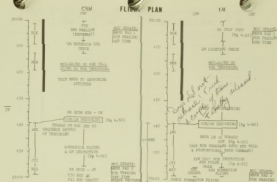


FLIGHT PLAN



FLIGHT PLAN

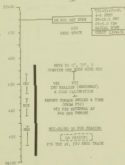


CSM
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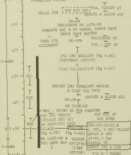
FLIGHT PLAN

TM

100



FORMATION FUTURE



MISSION	OPTION	DATE	TIME	REMARKS	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	01:00 - 04:00	1/20	1-12

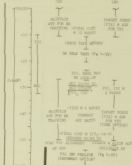
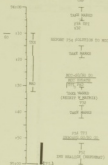
NO TIME FOR TOP 10

FLIGHT PLAN PAGE 1

CSM
000

FLIGHT PLAN

TM



MISSION	OPTION	DATE	TIME	REMARKS	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	04:00 - 07:00	1/20	1-12

NO TIME FOR TOP 10

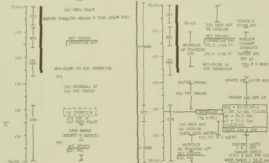
FLIGHT PLAN PAGE 2

CSM

FLIGHT PLAN

TM

1000



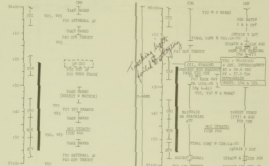
MISSION	ACTION	DATE	TIME	DAY/NIGHT	TIME
AFRIGO 9	FINC	February 3, 1960	05:00 - 06:00	5/00-05	7-64

104

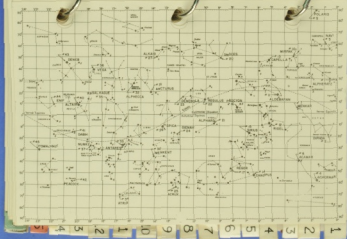
Filter PLAN

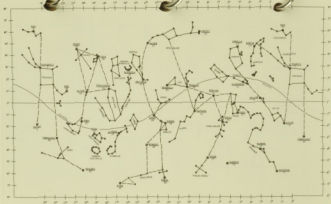
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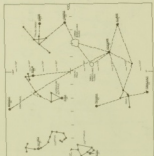
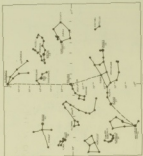
10



WISDOM	Source	Info	Time	Source	Ref
WISDOM 2	F2001	February 3, 1990	10:00 - 11:00	1/11	2-25







Basic Date
Changed
24, 1969

LM-3

RDZ-1

INTRAVEHICULAR TRANSFER TO LM

5 min

- 1
Slow In ISA:
SEQ Camera & Sun Lens
1 CEX360 Mag (L)
Camera Handle
CDR & LMP RDZ Charts & Procedures (2), &
Update Cards (May Be In LM)
EVA Checklist (1)

- 2
LMP Transfer To LM With LMP RDZ
Checklist & Flight Plan Card
Actuate CABIN DUMP VALVE & Open Hatch
EXTERIOR LTG - OFF
FLOOD LIGHT - ALL
DES H2O - OPEN

DOCKING TUNNEL INDEX ANGLE Δ

Transfer ISA To LM

ENTRY STATUS CHECK

10 min

- 1
CB(11)
Row 1&2: A11 Open
Row 3: HEATERS: RDZ RDR STBY - Close
LDG RDR - Close
LTG: ANJN/DOCK/COMPWT - Close
Row 4: PGNS: IMU STBY - Close
Row 5: EPS: BAT FEED TIE (2) - Close
CROSS TIE BAL LOADS - Close
DES ECA - Close
DFI PWR - OFF

FDI 1&2 - INRTL
EARTH/LUNAR - PWR OFF
LTG - OFF
MODE - HOLD/FAST
ALT SET - 130

FUEL & OXID VENT 4b-bp
MASTER ARM - OFF
- 3

ASC He SEL - BOTH
LDG GEAR DEPLOY tb-bp
STAGE-SAFE (Guarded)

4 S-BAND T/R - OFF (VOL-6)
ICS T/R - OFF (VOL-6)
RELAY - OFF
VOX-ICS (VOX SENS-7)
AUDIO CONT - NORM
VHF A/B - OFF (VOL-6)
MASTER VOL - 6
COAS - OFF

5 TTCA (COR) - JETS (On)

6 TIMER CONT - STOP
OVERRIDE ANUN - OFF
OVERRIDE NUM - OFF
OVERRIDE INTEGRAL - OFF
SIDE PANELS - OFF
FLOOD OVHD/PWD - BRIGHT
ANUN/NUM - DIM
INTEGRAL - DIM

7 X-POINTER SCALE - HI MULT
RATE/ERR MON - LDG RDR/CHPTR
ATTITUDE MON - PGNS
GUID CONT - PGNS
MODE SEL - LDG RADAR
RNG/ALT MON - ALT/ALT RT
SHFT/TRUN - +50°
RATE SCALE - 25"/SEC
THR CONT - AUTO
MAN THROT - COR
ENG ARM - OFF
X-TRANSL - 2 JETS
BAL CPL - ON
ASC He REG 1A2 - tb-gray
DESCENT He REG 1 tb-gray
DESCENT He REG 2 tb-bp
PRPLNT QTY MON - OFF
PRPLNT TEMP/PRESS MON - ASC
HELIUM MON - OFF
ABORT and ABORT STAGE - Flush/Guarded

Basic Date Feb. 24, 1969
Changed

Basic Date Feb. 24, 1969
Changed

LP

8 SYS A/B ASC FUEL & ASC OXID (4) - tb-bp
SYS A/B QUAD 1,4,2,3(R)-tb-gray
CRSPD - tb-bp
SYS A/B MAIN SOV - tb-gray
TEMP/PRESS MON - He
RCCA - OFF/RESET
RATE/ERR MON - LDG RDR/CHPTR
ATTITUDE MON - AGS
GLYCOL - PUMP 2
SUPT FAN - 1
O2/H2O QTY MON - ASC 2

9 ENG GMEL - ENABLE
DES ENG CMD OVRO - OFF
LDG ANT - DES
RADAR TEST - OFF
TEST MONITOR - ALT XMTR
SLEM RATE - HI
RNDZ RDR - AUTO TRACK
DEAD BAND - MIN
GYRO TEST - ROLL
ATTITUDE CONTROL (1) - MODE CONT
MODE CONTROL - ATT HOLD
DET - Up & STOP
TEMP MON - LDG
RCS SYS A/B-2 QUAD 1,2,3,4 - OFF
LTG: SIDE PANELS - OFF
FLOOD - ALL
OVHD/PWD - BRIGHT
EXTERIOR LTG - OFF
LAMP/TONE TEST - OFF
X-POINTER SCALE - HI MULT

10 ACA/4 JET (2) - ENABLE
TTCA/TRANSL (2) - ENABLE
RDC ANT RELEASE - UNSTOWED
AOT - CL, ANGLE - 0000

11 TTCA (LMP) - JETS (On)
AGS STATUS - OFF

12 CB(16)
ATT Open
Row 1: LTG: FLOOD - Close
Row 2:

- Row 3: STAB/CONT: ASA - Close
 Row 4: ALL Open
 HEATERS: 5-BD ANT - Close
 EPS: DES ECA - Close
 CROSS TIE BAL LOADS - Close
 BAT FEED TIE (2) - Close
- 13 POWER/TEMP MON - ED/OFF
 INVERTER - OFF
 BAT 1,2,3,4 - tb-bp
 DES BATS - tb-bp
 BAT 5&6 NORMAL & BACK UP FEED (4) tb-bp
- 14 AUDIO CONT - NORM
 S-BAND T/R - OFF (VOL-6)
 ICS T/R - OFF (VOL-6)
 RELAY - OFF
 VOX - ICS (VOX SENS-7)
 VHF A&B - OFF (VOL-6)
 MASTER VOL - 6
 VHF A RCVR VOICE BU - OFF
- 15 S-BAND MODULATE - PM
 XMTR/RCVR - OFF
 PWR AMPL - OFF
 VOICE - OFF
 PCM - OFF
 RANGE - OFF/RESET
 VHF A XMTR & RCVR - OFF (SQUELCH - 3)
 VHF B XMTR & RCVR - OFF (SQUELCH - 3)
 TELEMETRY BIOMED - OFF
 TELEMETRY - HI
 RECORDER - OFF tb-bp
 VHF - 1/PLSS TEST
 TRACK MODE - OFF
 PITCH - +255°
 YAW - +30°
 S BAND - 2
- 16 SUIT GAS DIVERter - PULL/EGRESS
 CABIN REPRESS - CLOSE
 PLSS FILL - CLOSE
 PRESS REG A&B - CLOSE
 DES O2 - CLOSE

Feb. 24, 1969
 Basic Date -
 Changed

Feb. 24, 1969
 Basic Date -
 Changed

LM-3

- T&2 ASC O2 - CLOSE
 SUIT ISOL (2) - SUIT DISC
 SUIT CIRCUIT RELIEF - AUTO
 CABIN GAS RETURN - AUTO
 CO2 CANISTER SEL - PRIM
 PRIM & SEC CO2 CANISTER - CLOSE
 WATER SEP SEL - PULL/SEP 2
 ASC H2O - CLOSE
 SEC EVAP FLOW - CLOSE
 PRIM EVAP FLOW No. 2 - CLOSE
 DES H2O - OPEN
 WATER TANK SELECT - DES
 SUIT TEMP - COLD
 CABIN TEMP - NORM
- CABIN RELIEF AND DUMP (2) - AUTO
- DFI PRIMARY - ON, SECONDARY - OFF
 UTILITY LIGHTS (Both) - As Required
 Fwd Hatch Closed & Locked

EPS ACTIVATION & CHECKOUT

5 min

- CSM Position LM PWR - RESET Then OFF
 LTG: ANUN/NUM - BRIGHT (1 Caution, 9
 Power Failure Lts - On)
- CB(11) INST: SIG COND 1 - Close
 EPS: XLUNAR BUS TIE - Close
 DES ECA CONT - Close
 DC BUS VOLT - Close
- CB(16) INST: SIG SENSOR - Close
 PCM/TE - Close
 SIG COND 2 - Close
 COMM: PRIM S-BD PWR AMPL - Close
 PRIM S-BD XMTR/RCVR - Close
 PMP - Close
 EPS: DISP - Close
 DC BUS VOLT - Close
 DES ECA CONT - Close
 XLUNAR BUS TIE - Close

- 3 CB(11) COMM: VHF B XMTR - Close
VHF B - DATA
TELEMETRY - LO
CSM Record LBR Data
- 4 Verify BAT 1,2,3,4 - tb-LO
DES BATS - gray
BATS S&M NORMAL & BACKUP (4) tb-bp
Check BAT And BUS Voltages (When BUS
Voltages Less Than 27v Select
High Voltage Taps)
- CB(16)EPS:CROSS TIE BAL
LOADS-open
BAT 1 HI VOLTAGE-OFF/RESET
tb-bp
BAT 1 HI VOLTAGE - ON
tb-gray
Repeat For BATS 2,3,4
CB(16)EPS:CROSS TIE BAL
LOADS-Close
- 5 CB(11) AC BUS BMA: BUS TIE INV 2&1 (4)
- Close
AC BUS VOLT - Close
EPS: INV 1 - Close
CB(16) EPS: INV 2 - Close
- 6 POWER/TEMP MON - AC BUS
INV - 1 Then 2
CB(11) EPS: INV 1 - open

ACTIVATE S-BAND FOR TM

1 min

- 1 S-BAND-PH, PRIM, PRIM, VOICE, POM,
RANGE, OFF, LO
S-BAND ANTENNA - As Desired

GLYCOL PUMP ACTIVATION

1 min

- CB(16) ECS: DISP - Close
CB(11) ECS: GLYCOL PUMP AUTO TRNFR-Close
GLYCOL PUMP 1 - Close
GLYCOL PUMP AUTO TRNFR - Open
GLYCOL - PUMP 1
CB(11) ECS: GLYCOL PUMP 2 - Close
GLYCOL PUMP AUTO TRNFR - Close

CAUTION/WARNING CHECKOUT

2 min

- CB(16) INST: CMEA - Close (LOC, CES
AC, CES DC Warning Lts, PRE AMPS,
HEATER, GLYCOL, ECS Caution Lts,
H2O SEP Comp Lts-On)
LTG: ANUN/DOCK/COMBNT - Close
MASTER ALARM - Close (Master Alarm - On)
HEATERS: DISP - Close

- RCS TEMP/PRESS MON - Cycle Then He
HTR TEMP MONITOR - Cycle Then LOG
DRO TEST - POS RT
LAMP/TONE TEST - Check All Positions

Basic Date Changed Feb. 24, 1969

Basic Date Changed Feb. 24, 1969

CIRCUIT BREAKER ACTIVATION

5 min

24, 1969
Basic Date
Changed — 25, 1969

LM-3

- 1
Row 1: CB(11)
All Closed Except:
AC BUS B&A: AOT LAMP (2) - Open
RNDZ RDR - Open
Row 2: AC BUS A: RNS/RNG RT ALT/ALT RT - Close
All Closed Except:
RCS SYS A: QUAD 4,3,2,1 TCA (4) - Open
Row 3: All Closed Except:
STAB/CONT: ABORT STAGE - Open
(Possible Master Alarm-On With ATCA [PGNS])
AELD - Open
ATT DIR CONT - Open
ED: LDG GEAR FLAG - Open
Row 4: All Closed Except:
ECS: SUIT FAN 1 - Open
PGNS: LDG RDR - Open
RNDZ RDR - Open
(Possible LGC Warning and RESTART
Lt - On With LGC/DSKY)
IMJ OPR - Open
Row 5: All Closed Except:
EPS: CROSS TIE BUS - Open
ASC ECA CONT - Open
INV 1 - Open
SLA/LEM ANT TRNFR (2) - Open
R/D INST A&B - Open
DFI PWR - OFF
- 2
Row 1: CB(16)
All Closed Except:
RCS: SYS B QUAD 1,2,3,4 TCA - Open
Row 2: All Closed Except:
STAB/CONT: AEA - Open
AELD - Open
ABORT STAGE - Open
Row 3: All Closed Except:
COMM: TV - Open
ECS: GLYCOL PUMP SEC - Open
(Suit Fan Comp Lt-On With Suit Fan 2P
& Master Alarm May Go On With CO2
SENSOR)
Row 4: All Closed Except:
CAMR: SEQ - Open

EPS: ASC ECA CONT - Open
CROSS TIE BUS - Open

MISSION TIMER - SET

PRIM EVAP FLOW - OPEN (When Glycol
Temp >60°F)

RES SYS A/B-2 RUAD 1,2,3,4 - AUTO

CB(16) INST: CWEA - Open & Reclose (Master
Alarm, WEATER, ECS, GLYCOL Caution, H2O
SEP, And SUIT FAN Lts - On)

TR VERIFICATION

2 min

FUEL & OXID VENT (2) - bp
LOG GEAR DEPLOY - bp

ASCENT He REG 1&2 - gray
DESCENT He REG 1 - bp
DESCENT He REG 2 - bp

SYS A&B ASC FUEL & OXID (4) - bp
SYS A&B RUAD 1,4,2,3(8) - gray
CRSFD - bp
SYS A&B MAIN SOW - gray

TAPE - bp

ECS ACTIVATION & CHECKOUT

5 min

O2/H2O QTY MON - ASC 2, ASC 1, DES

SUIT ISOL (Both) - SUIT FLOW
SUIT ISOL (Both) - ACTUATE OVRD (SUIT
Discon)

DES O2 - OPEN
SUIT GAS DIVERTER - PUSH/CABIN
PRESS REG A&B - EGRESS (SUIT Gas Divertor
Automatically Extends & Cabin Fans
Go Off)

SUIT PRESS 4.8 To 5.2 psia ~~4.8 To 5.2~~
CB(16) ECS: CABIN FAN CONT - Open
(Cabin Fans Go On)

SUIT FAN - 2 (Master Alarm, SUIT/FAN
Warning Lt-On momentarily, ECS Caution
Lt, H2O Sep Comp Lt - On Then Off in
<2min)
PART PRESS CO2 <7.6 mm hg

SUIT FAN/H2O SEP CHECK

2 min

CB(16) ECS: SUIT FAN 2 - Open (Master
Alarm, SUIT/FAN Warning Lt, Suit Fan
Comp Lt-On)

When ECS Caution Lt & H2O SEP Lt-On
Master Alarm - On

CB(11) ECS: SUIT FAN 1 - Close
H2O SEP SEL - PUSH/SEP 1

SUIT FAN - 1 (SUIT/FAN Warning Lt, Suit
Fan Comp Lt - Out, ECS Caution Lt,
H2O Sep Comp Lt - Out in <2 min)
CB(16) ECS: SUIT FAN 2 - Close

S-BAND AND VHF ACTIVATION

Set Comm Configuration
S-Band - PW, PRIM, PRIM, VOICE, PCM, RANGE, L/R, L
VHF - ON, ON, VOICE, ON

S-BAND & VHF ANTENNA - As Desired

AUDIO (Both):

S-BAND - T/R (Vol To Just Hear MSFN)

ICS - T/R

RELAY - OFF

AUDIO CONT - NORM

VOX - ICS

VHF A (CDR) - T/R

(LMP) - OFF

VHF B (CDR) - OFF

(LMP) - T/R

Connect Omphical To Right Side
(Red/Red - Blue/Blue)

Basic Date
Changed

24, 1969

Basic Date
Changed

24, 1969

LM-3

LMP
PIT &
VHF
INOP

RDZ-11

CSM Configure Simplex B And Set LMP
 Suit Pwr To Off
 Connect To LM Comm Umbilical (Audio, Biomed)
 TAPE - ON
 Conduct COMM Check, Adjust Squelch (ACA & Umbilical PTT)

AUDIO (LMP): VOX - ICS/XMTR
 Conduct Comm Check (Adjust VOX SENS)
 VOX - ICS

CSM Configure For Simplex A
 AUDIO (LMP): VHF A - T/R
 VHF B - OFF
 Conduct COMM Check, Adjust Squelch

CSM Configure For LM Data
 VHF B XMTR - DATA
 VHF B RCVR - OFF
 TELEMETRY - LO

VHF ANTENNA - Select Other Antenna
 Perform COMM Check With CSM

CSM Select Other Antenna
 Perform COMM Check With CSM

VHF ANTENNA - Select Original Antenna
 Conduct COMM Check
 TAPE - OFF

Select Best Antenna Combination

Basic Date - Feb. 24, 1969
 Changed

Changed

Basic Date - Feb. 24, 1969
 Changed

LM-3

RDZ-12

CDR TRANSFER & CREW CONNECT

10 min

CSM Set LMP Suit Flow Control To Off
 LMP SUIT ISOL - SUIT FLOW (Verify Flow)
 Disconnect Umbilical And Pass To CSM
 Receive And Stow ISA
 LMP Configure Cabin (Straps, Utility
 Lights, Cards, Restraints, etc.)

CDR Transfer To LM With CDR RDZ Checklist
 Connect To CDR Hoses (Red/Red, Blue/Blue)
 CDR SUIT ISOL - SUIT FLOW
 Connect To LM Comm Umbilical (Audio, Biomed)
 Conduct Comm Check (ACA & Umbilical PTT)

CDR AUDIO: VOX - ICS/XMTR
 Conduct Comm Check (Adjust VOX SENS)

CDR AUDIO: VOX - ICS

OPS CHECK

CDR HTR CHECK NO GOOD
 5 min

OPS O2 PRESS 6380 To 6380 psi
 O2 HOSE - LOCKED
 OPS O2 - ON (O2 Press 3.4 To 4.0 psi)
 Verify O2 Flow
 HEATER TEST BUTTON - DEPRESS
 (At Least 1 Lt - On)
 OPS O2 - OFF
 Verify REG Press Gauge Reads Zero
 In About 4 min

STOW OPS (Both On Pallet)

DROGUE AND PROBE INSTALLATION

10 min

Both Electrical Umbilicals Disconnect,
 And Secured
 Drogue Lock Lever Engaged And Flush
 Three Capture Latches Engaged And Locked
 (Before & After Preload)

LN Hatch Exterior Insulation OK, Flaps Secured

Close & Secure Hatch
CABIN DUMP VALVE - AUTO
CABIN REPRESS - AUTO
PRESS REG A/B - CABIN
SUIT GAS DIVERter - PUSH/CABIN
CB(16) ECS: CABIN FAN CONT - Close

Basic Date Feb. 24, 1969
Changed
Changed

Basic Date Feb. 24, 1969
Changed
Changed

LN-3

GLYCOL PUMP CHECK

5 min

CB(11) ECS: GLYCOL PUMP 1 - Open (Master Alarm, And ECS Caution Lt On Momentarily)
CB(11) ECS: GLYCOL PUMP 1 - Close,
[Check Glycol Comp Lt-On]

GLYCOL - INST(SEC) (Note Glycol Press)
CB(14) ECS: GLYCOL PUMP SEC - Close
(10-20 psi Rise) ~~10-20~~ - 15-30 psi
ECS: GLYCOL PUMP SEC - Open

GLYCOL - PUMP 2 (15-30 psi)
CB(11) ECS: GLYCOL PUMP AUTO TRNFR - Open
GLYCOL - PUMP 1 (Master Alarm & ECS
Caution Lt On Momentarily, 15-30 psi)
CB(11) ECS: GLYCOL PUMP AUTO TRNFR - Close

AIR/PGA PRESSURE INTEGRITY CHECK

5 min

SUIT GAS DIVERter - PULL/EGRESS
CABIN GAS RETURN - EGRESS
SUIT CIRCUIT RELIEF - CLOSE
PRESS REG A - CLOSE
PRESS REG B - DIRECT O2 (Suit Press To 9.85 psia)
PRESS REG B - CLOSE (Monitor Cuff Gage
Decay <.3 psi in 1 min)

CO2 CANISTER SEL - SECONDARY (CO2
Comp Lt - On Monitor Cuff Gage
Decay at <.3 psi in 1 min)

CO2 CANISTER SEL-PRIMARY(CO2 Comp Lt-Off)
SUIT CIRCUIT RELIEF - AUTO
PRESS REG A/B - CABIN
CABIN GAS RETURN - AUTO
SUIT GAS DIVERter - PUSH/CABIN

WIP NOT
ON LOUP
DUE TO
RIGHT
EAR NOT
CLEARING

ASCENT BATTERY ACTIVATION & CHECKOUT 2 min

CB(16) EPS: ASC ECA CONT - Close
CROSS TIE BAL LOADS - Open

POWER/TEMP MON - BAT 5
BAT 5 NORMAL FEED - ON tb-gray
Verify BAT Current Then BAT & SE
BUS Voltage

BAT 1,2 HI VOLT - OFF/RESET tb-bp
(Possible Master Alarm and DC BUS
Warning Lt - On)

POWER/TEMP MON - BAT 6
BAT 6 BACKUP FEED - ON tb-gray
Verify BAT Current Then
BAT & SE BUS Voltage

BAT 5 NORMAL FEED - OFF/RESET tb-bp

POWER/TEMP MON - BAT 1
BAT 1,2, HI VOLT - ON
BAT 6 BACKUP FEED - OFF/RESET tb-bp

POWER/TEMP MON - BAT 6
BAT 6 NORMAL FEED - ON tb-gray
Verify BAT Current Then BAT & COR
BUS Voltage

BAT 3,4 HI VOLT - OFF/RESET tb-bp
(Possible Master Alarm And DC BUS
Warning Lt - On)

POWER/TEMP MON - BAT 5
BAT 5 BACKUP FEED - ON tb-gray
Verify BAT Current Then BAT & COR
BUS Voltage

BAT 6 NORMAL FEED - OFF/RESET tb-bp

POWER/TEMP MON - BAT 4
BAT 3, 4 HI VOLT - ON tb-gray
BAT 5 BACKUP FEED - OFF/RESET tb-bp

CB(16) EPS: ASC ECA CONT - Open
CROSS TIE BAL LOADS - Close

Basic Date
Changed
24, 1969

ASCENT BATTERY ACTIVATION & CHECKOUT

1 CB(16) EPS: ASC ECA CONT - Close

2 POWER/TEMP MON - BAT 5
BAT 5 NORMAL FEED - ON tb-Gray
Verify Bat Current

3 POWER/TEMP MON - BAT 6
BAT 6 NORMAL FEED - ON tb-Gray
Verify Bat Current

4 BAT 1,2,3,4 HI VOLT - OFF/RESET tb(4)-bp
Verify Bats 1,2,3,4 Current Zero

5 BAT 5,6 BACKUP FEED - ON tb(2)-Gray
BAT 5,6 NORMAL FEED - OFF/RESET tb(2)-bp
Verify Bat 5,6 Current

6 BAT 1,2,3,4 HI VOLT - ON tb(4)-Gray
Verify Bat 1,2,3,4, Current
BAT 5,6, BACKUP FEED - OFF/RESET tb(2)-bp
Verify Bat 5,6 Current Zero

7 CB(16) EPS: ASC ECA CONT - Open

8 REPORT ED BAT VOLTAGE TO MSFN
BAT A 16.8 BAT B 37.5

BUSSES 29.0 VDC
ON ASC BATTs

5

6

7

8

9

10

11

12

13

14

5

6

PGNS TURN ON & SELF TEST

IF STBY Lt-On, PRO

V36E

V21N01E, 3000E, 2176E, E
3011E, 201E, E
1642E, 3777E

V66E

CB(11) PGNS IMU OPR - Close (NO ATT
Lt-On For 90 sec)

V36E

F 88 88

DSKY LIGHT CHECK

(Master Alarm, LSC, ISS Warning
And All DSKY Lights - On For 5 sec)
Key RSET
When NO ATT - Off +20 sec, V37E00E

V25 N01E, 1365E

F 21 01

E, E, E

V16 N01E, 1365E

15 01

R1, R2, R3 All Zero

V21 N27E, 10E INITIATE SELF TEST

KEY REL

15 01

R1 Number Of Errors

R2 Number Of Tests Started

R3 Number Of Tests Successful

(Test Successful When R2 > 3)(78 sec)

*PROG Lt - On

*V05N09E 01102 Self Test Error

*N00E Record For MSFN

* R1

* R2

* R3

V21 N27E 0E TERMINATE TEST

Basic Date
Changed
For 24, 1969

18-3
LP-3

ON LOG/DSKY
CB-LOG WE

5 min
WERE NOT IN
STBY- INSTEAD
F37 P06

RDZ-18

AGS ACTIVATION, SELF TEST

2 min

AGS STATUS - STBY
 CB(16)STAR/CONT: AEA - Close
 AGS STATUS - OPERATE
 (Master Alarms & AGS Warning Lt - On,
 Then Off)

6666 (OPR ERR - On)

000+88888

123 - 45679

412R + 1 SELF TEST SATISFACTORY
 + 3 LOGIC TEST FAILURE
 + 4 MEMORY TEST FAILURE
 + 7 LOGIC AND MEMORY TEST FAILURE
 (To Reinitiate Test Set 412 + 0)

574R (+) DESCENT STAGE FLAG (+Not Staged)

604R(+) LUNAR SURFACE FLAG (+Not On
Lunar Surface)612R(+0000) STAGING SEQ COUNTER (+00000
For Att Hold At Abort Stage)

RATE GYRO CHECK

88.52

2 min

RATE SCALE - 25°/SEC
 GYRO TEST - POS RT (Roll Rate +5°/sec)
 GYRO TEST - NEG RT (Roll Rate -5°/sec)
 GYRO TEST - PITCH (Repeat Test)
 GYRO TEST - YAW (Repeat Test)

RATE SCALE - 5°/SEC
 Repeat Tests

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LM-3

90:32 - 90:38

RDZ-19

**[LM DOCKED IMU] ALIGNMENT

7 min

Verify CSM In MIN DB ATT HOLD Until
 Coarse Align Complete

Calculate LM Gimbal Angles:

OG	IG	MG
300.00	090.00°	
+ 2.10		
302.36		360.00
- 147.49	+ 290.00	- 346.71
+ 154.87	+ 420.00	+ 13.29

F 21 22 V41 N20E COARSE ALIGN IMU
 LOAD 1CDU ANGLES OG, IG, MG (.01°)
 (NO ATT Lt - On, F0AI Torques)
 *PRG Lt - on *
 *V05N09E 00211 COARSE *
 * ALIGN ERROR, GO *
 * TO 3 *

V40 N20E ZERO CDU (NO ATT Lt - Off)
 Notify CSM ATT HOLD No Longer Required

F 21 07 V25 N07E
 SET REFSMFLG
 77E, 10000E, 1E
 V01 N01E, 77E, Confirm Bit 13 Set

V37E51E, PR0, V37E00E

ROZ-20

06 20 V06 N20
On CSM MARK - ENTR
Copy OG, IG, MG, CSM & LM

OG	IG	MG
147.35 CM	289.80 CM	346.53 CM
154.76 LM	019.07 LM	013.05 LM

Voice Angles To MSFN

Copy Ground Calculated Gyro Torquing Angles

91.06 X-06379, Y-00790, Z-00310

9 F 21 93 V42E FINE ALIGN IMU 3 min
LOAD GYRO TORQUING ANGLES X,Y,Z (.001")

10 F 16 93 V16 V93E
MONITOR TORQUING

LGC CLOCK INITIALIZATION 8 min

1 5064 V37E OGE 90'00:00
2 5064 V25 N36E Load CSM Time But Do Not ENTR
-16 Seconds, On CSM MARK - ENTR

3 0188 V06 N65 ON CSM MARK - ENTR
0611 COMPUTE CSM/LM & TIME
-23 Perform Several Times Then

4 1037 V55E LOAD & TIME
1051
-14

✓ 187 TERPHEN 90'03:00 2 min

1 00007 CSM V05 NOTE, 1706E And Read TERPHEN

2 11153 V25 NOTE, 1706E (Load TERPHEN) E

3 V05 NOTE, 1706E Verify TERPHEN

ROZ-21

✓ OPS RCS TEMP/PRESS CHECK
RPS PRESSURIZATION 90:20

4 min

PRPLNT QTY MON - DES 1, DES 2, Then OFF
(Reading Not Valid)

PRPLNT TEMP/PRESS MON
DES 2&1 (50°-75°F, 238-253 psi)
ASC (50°-90°F, 80-200 psi)

3 HELIUM MON
DES AMB PRESS (238-253 psi)
DES SUPCRIT PRESS (1320-1435 psi)
ASC TEMP 1&2 (30°-140°F)
ASC PRESS 1&2 (2720-3500 psi)
OFF

4 ASCENT He REG 1&2-Open tb(2)-gray
ASC He SEL - BOTH
MASTER ARM - ON
ASC He PRESS - FINE
MASTER ARM - OFF (Master Alarm - On)

5 PRPLNT TEMP/PRESS MON 48-60 190-190
ASC TEMP/PRESS (50°-90°/172-203 psi)
DES 1 63-63 2840-2820-14C

6 RCS QUANTITY AMB 98-77
RCS TEMP/PRESS MON: 2800-2780 67-66
He (2475-3225 psi)
PRPLNT (40°-100°F) (178-192 psi) 180-180
FUEL MANF (175-189 psi) 162-180
OXID MANF (175-189 psi) 162-181
He 00-03

7 HTR: RCS TEMP MONITOR - QUAD 1,2,3,4
(120°-190°F)

8 RCS SYS AMB ASCENT FEED 2 (2) - CLOSE
RCS SYS AMB ASCENT FEED 1 (2) - OPEN
ASC FUEL & ASC OXID tb(4) Remain bp

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40131 ✓

RDZ-22

BOS COLD FIRE

R min ~ 4

1 CSM MIN DEAD BAND And ATT HOLD
 2 X-TRANSL - 4 JETS
 3 Verify QUAD TEMP ~~120°F~~
 4 SYS AAR NDAO ISOL (R)-OPEN to-gray
 5 RT115UDE CONTROL (3)-MODE CONT
 6 MODE CONTROL-ATT HOLD

2 V40E LOAD GAP, 02012E

3 V77E
 4 V15 N10E, 42E
 5 CDR ACA (To Soft Stop)
 6 Roll Right - R1 00051
 7 Roll Left - 77726
 Pitch Up - R1 00051
 Pitch Down - 77726
 Yaw Right - R2 77726
 Yaw Left - 00051

4 LMP Repeat Test

5 V11 N10E, 5E
 6 CDR ACA
 7 Roll Right - R1 00245
 Roll Left - 00132
 Pitch Up - R1 00226
 Pitch Down - 00151
 CDR TTCA
 Up - R1 00250
 Down - 00125

6 LMP Repeat Test

7 E6E
 CDR ACA
 Yaw Right - R1 00250
 Yaw Left - 00125
 CDR TTCA
 Right - R1 00220
 Left - 00140
 Yaw - R1 00011
 Att - 00006

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RDZ-23

LMP Repeat Test

V76E, V11 N10E, 31E R1 67777

CDR ACA

Roll Right - R1 27757
 Roll Left - 27737
 Pitch Up - R1 27776
 Pitch Down - 27775
 Yaw Right - R1 27767
 Yaw Left - 27773

LMP Repeat Test

MODE CONTROL - AUTO R1 57777

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W-3 Basic Date 8-24-1969

027-74

E RASABLE MEMORY DUMP

CB(11) R/D INST B - Close

Verify 95% Contact.

V70E (42 sec + 2 Dumps)

**REFORMAT AND STATE VECTOR UPDATE

Y37E 00H

MSFN DoLink REFSMART & State Vector And

Nav Check Pad

(Unlink Acty Lt - On Then Off When

UsLink Completed)

*GINGIVAL DRIVE AND THROTTLE TEST

4. $\frac{1}{2}$ mile.

TWR CONT = MAN

MAN THROT = SE

TTCA (Bath) = THROTTLE(UP)MIN POSITION

Verify MSFN Contact

V49E

F 01 46

DAP CONFIGURATION

R1 02012

PRO

\$ 06.47

LM, VT, COM, WT

81

R2 27007

PRO

† D6 47

ENGINE CLIMBQ, TRIM, PITCH, ROLL (1.00°)

R1 = 00428 (uncert) (+00458)

82 40730 40730 (40730)

Verify MODE CONTROL - AUTO

ENGINE GIMEL - ENABLE

ENGINE ARM - DES

PRO (Master Alarm, GDA/SCCA Warning Lt-On)

When Cymbal's Reach Limits)

5 TTCA (LMP) - MIN, Then SOFT STOP (51s)
Then MAX (Off Scale High) Then MIN
MAN THROT - CDR
TTCA (CDR) - MIN, Then SOFT STOP
(51s) Then MAX (Off Scale High) Then MIN

6 THR CONT - AUTO

7 F 50 48 TRIM COMPLETE
ENG ARM - OFF (GDA/RCCA Warning Lt-Off)
TTCA (Both) - JETS (DN)
PRO
MSFN Verifies Final GDA Position; If
Gimbal Angles Differ From Desired
Values By More Than 0.1" Repeat V48

8 CB(11) STAB/CONT: DECA PWR - Open
CB(16) STAB/CONT: DES ENG OVRO - Open

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✓ **RCS HOT FIRE

91:18 - 91:21 5 min

1 Notify CSM
Verify HBR With MSFN
Verify RCS QUAD TEMP 1,2,3,4 120°-190°F
For At Least 25 min
CB(16) INST: CWEA - Open & Reclose
(Master Alarm - On)
GUID CONT - AGS
X-TRANSL - 4 JETS
DEAD BAND - MAX
ATTITUDE CONTROL (3) - MODE CONT
MODE CONTROL - ATT HOLD

2 CDR ACA (Deflect Slowly To Soft Stop,
Pause At Null)
Roll Right, Left, Pitch Up, Down, Yaw
Right, Left. (Possible Master Alarm,
RCS TCA Caution Lt - On tb - red)
Verify MSFN Read RCS Data

ATTITUDE CONTROL (3) - PULSE
CB(11) STAB/CONT: ATT DIR CONT - Close
CDR ACA (Deflect Hardover, Pause at Null)
Roll Right, Left, Pitch Up, Down, Yaw
Right, Left (Possible Master Alarm,
RCS TCA Caution Lt - On, tb - red)

4 CB(11) RCS: SYS A QUAD 4,3,2,1 - Close
CB(16) RCS: SYS B QUAD 1,2,3,4 - Close
CB(16) INST: CWEA - Open & Reclose
CDR TTCA
Up, Down, Right, Left, Fwd, Aft

5 V76E
GUID CONT - PGNS
CDR TTCA (Pause at Null)
Up, Down, Right, Left, Fwd, Aft
Standby For MSFN GO

6 X-TRANSL - 2 JETS
DEAD BAND - MIN

7 Set DAP - 0100Z

9

10

11

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6

90:46-90:52

R02-27

REGULATOR CHECK

10 min

Verify CSM Tunnel Hatch Is Closed,
Tunnel Depressed

CABIN GAS RETURN - EGRESS

CABIN REPRESS - AUTO

PRESS REG B - EGRESS (Cabin Fans - Off
And Suit Gas Diverter - Egress)

FWD CABIN DUMP VALVE - OPEN Then AUTO At
4.0psi (Master Alarm, CABIN Warning Lt
And Auto Cabin Repress - On At 4.45 psi)

As Soon As Cabin Repress Starts

PRESS REG A - CLOSE

(CABIN Warning Lt - Off, Cabin Repress
Stops)

CABIN REPRESS - CLOSE

FWD CABIN DUMP VALVE - OPEN Then AUTO At
3.5 psi (Verify Suit Press 3.6 To 4.0 psi)

PRESS REG B - CLOSE (Possible Master Alarm,
CABIN Warning Lt On Momentarily &
Cabin Fans-On)

SUIT CIRCUIT RELIEF - OPEN Then AUTO At
Suit Press of 3.5 psi

PRESS REG B - EGRESS (Suit Press 3.6 To
4.0 psi Master Alarm & CABIN Warning
Lt - On Momentarily, Cabin Fans - Off)

PRESS REG B - CLOSE (Master Alarm, CABIN
Warning Lt On Momentarily & Cabin
Fans - On)

SUIT CIRCUIT RELIEF - OPEN Then AUTO At
Suit Press of 3.5 psi

PRESS REG A - EGRESS (Suit Press 3.6 To
4.0 psi Master Alarm, CABIN Warning Lt -
On Momentarily, Cabin Fans - Off)

CABIN REPRESS - AUTO

PRESS REG B - CABIN (Master Alarm, CABIN
Warning Lt, Cabin Fans-On, Suit & Cabin
Press Rises To 4.6 To 5.0 psi)

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LM

PRESS REG A - CABIN
CABIN GAS RETURN - AUTO
SUIT GAS DIVERTER - PUSH/CABIN

PIPA BIAS CHECK 90:44 - 90:45 3 min

DET - ZERO
Request CSM Null Bytes <1°/sec
Request CSM Go CMC MODE - FREE

V25N21E, E, E, E/Start DET

YD6E

06 21 XYZ PIPA PULSES (+pulses)

At T+32sec - ENTR

T+32sec [X]R1 + ~~09~~ (+XXXAB)

(Y)R2 + 0

(Z)R3 + 0

Notify CSM Test Complete

V06N01E, 1452E (R1-Review X Bias)E (+AB000)

1454E (Review Y Bias)E

1456E (Review Z Bias)

V21 N01E

F 21 01 LOAD 1452E (Calculated X BIAS)E, E (+AB000) + 12

1454E (Calculated Y BIAS)E

1456E (Calculated Z BIAS)E

(Use Same Sign As In Measured Bias
of Step 4)

AGS INITIALIZATION

3 min

AGS In Operate for 25 min
SET AGS TIME (BIAS 90 hr.)
UPDATE AGS
ALIGN AGS
Check 403(Hp) & 315(Ha) With YD2E

AGS CALIBRATION

9 min

Read And Record

540R X ACCEL BIAS COEFF + 0 (Octal)

541R Y ACCEL BIAS COEFF + 0 (Octal)

542R Z ACCEL BIAS COEFF - 7777 (Octal)

544R X GYRO DRIFT COEFF + 0 (.01°/hr)

545R Y GYRO DRIFT COEFF + 0 (.01°/hr)

546R Z GYRO DRIFT COEFF + 0 (.01°/hr)

Verify AGS In Standby/Operate For 25 min,
PGNS-ON, LM Thrusters Disabled,
Rates <1°/sec, RPY ICDS Torqued
Beyond 11.25° And Will Not Pass Thru 0°
45°, 90°, etc. (CSM 06-02.5°, 16-22.5°,
MG-22.5° Will Give The Starting Attitude)

CSM Establish AGS Calibration Attitude,
Minimize Rates, Go CMC MODE - FREE

V16N20E Monitor ICDU Angles (All Angles
should be Approx. 22°, 67°, 112°, 157°,
202°, 247°, 292°, or 337°)

LM ICDU's: 000 112.5°

202.5°

022.5°

FDAI ANGLES: R 132.7°

P 339.8°

Y 301.4°

V40 N20E ZERO CDU

400 + 6E CALIBRATE GYRO & ACCEL

Read And Record After 32Sec

540R X ACCEL BIAS COEFF + 0 (Octal)

541R Y ACCEL BIAS COEFF + 0 (Octal)

542R Z ACCEL BIAS COEFF - 7777 (Octal)

(If BIAS Changes > 4 Counts, AGS Failed)

If It Appears that The Gimbal Angles Will
Pass Thru 0°, 45°, 90°, 135°, 180°,
225°, 270°, or 325°. Exit Calibration By
400+6E

400R + 0 when GYRO & ACCEL CALIBRATE COM-
LETE

Read And Record After 5 min 2 sec

SAVR X GYRO DRIFT COEFF ± 1 (.01"/hr)SAVR Y GYRO DRIFT COEFF ± 1 (.01"/hr)SAVR Z GYRO DRIFT COEFF ± 1 (.01"/hr)

(If GYRO DRIFT > 2.50"/hr, AGS Failed)

CHANGES

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LANDING RADAR SELF TEST

1 X-POINTER (Both) - HI MULT
RATE/ERR MON-LDG RDR/CMPT4
TEMP MON - LDG RDR (>49°F) =====
RNG/ALT MON - ALT/ALT RT
LDG ANT - DES
MODE SEL - LDG RDR
CB(11) PGNS: LDG RDR - Close
(X-POINTER Will Oscillate Then
Up And Right Off Scale)

2 RADAR TEST - LDG (Alt And Alt Rt Tapes
Drive)
TEST MONITOR-ALT XMTR(2.1 To 5.0V)(3.6V) 2.7
-VEL XMTR(2.1 To 5.0V)(3.8V) 3.75
ALT/ALT RT MON - +8094 To +8457 ft/-433 To
-465 fps (8280/-450) 8470/-462.5
LDG ANT - MONTR (10 sec)
ALT/ALT RT+7818 To +8169 ft/-441 To -457 fps
(8000/-450) 8000/-462.5

LDG ANT - DES (Wait 10 sec)

V62E INITIATE RDR SELF TEST

F 04 06 R1 00004 SPECIFY RDR
R2 00002 LDG RDR
PRO

6 F 16 66 SLANT RANGE, ANT POSITION (ft)
R1 +08165 To +08418 (88286) 8272
R2 +00001
PRO $\pm$

7 F 16 67 LDG RDR VEL X, Y, Z (fps)
R1 +00230 To -00264 (-00246) -247
R2 +00924 To +00954 (+00930) 7930
R3 +00643 To +00689 (+00666) 7667

8 V34E

9 LDG ANT - AUTO

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10 V61 COMMAND ANT TO POS 2 (27 sec)
ALT/ALT RT MON +7818 To +8169ft/-441 To
-457 fps (8000/-450) **8100/-450.5**

11 F 04 06 V62E INITIATE RDR SELF TEST
R1 00004 SPECIFY RADAR **8281**
R2 00002 LDG RDR **+2**
PRO

12 F 16 66 SLANT RNG, ANT POSITION (ft) **7905**
R1 + 08156 To +08418 (08292)
R2 00002 **8000**
V34E **-450.5**

13 RADAR TEST - OFF
CB(11) PGMS: LDG RDR - Open
(Master Alarm - On)

✓ RNDZ RDR SELF TEST
AND SET FOR UNDOCKING

10 min

1 Verify CSM RCS Thruster 83 And Transponder
OFF
RDZ ANT - Released
X-POINTERS (Both) - HI MULT
RATE/ERR MON (Both) - RNDZ RADAR
ATTITUDE MON (Both) - PGMS
MODE SEL - LDG RADAR
RNG/ALT MON - RNG/RNG RATE
SHFT/TRUN - +50°
RDZ RDR - SLEW
TEMP MONITOR - RNDZ (+10° To 150°)

2 CB(11) AC BUS A: RNDZ RDR - Close (Wait
30 sec)
PGMS: RNDZ RDR - Close (NO TRACK Lt - On)

3 SLEW LEFT TO 0°, 0°
SLEW RATE - LO
SHFT/TRUN +5°
Slew Antenna Up, Dn, Left, Right To
Verify Slew

4 RDZ RDR - AUTO TRACK (Master Alarm & RNDZ
RDR Caut Lt - On)
RADAR TEST - RNDZ (Rng Rt. Tape Drives,
X-Pointers And FDI) Needles Vary Between
Limits. After 12sec, Rng Tape Drives,
NO TRACK And RNDZ RDR Caut Lt - Out)

5 TEST/MONITOR-AGE 0.7 To 3.5v(1.5)(**2.6**)
-XMTX PWR 2.1 To 4.8v(2.8)(**3.1**)
-SHAFT ERR 1.5 To 3.5v(**2.2-2.6**)
-TRUN ERR 1.5 To 3.5v(**2.2-2.5**)
-AGE

RDZ RDR - SLEW
Slew Antenna To 0°, 0°
RDZ RDR - LGC (NO TRACK Lt - On)

V62E START RDR SELF TEST
R1 00004 SPECIFY RADAR
R2 00001 RNDZ RADAR
PRO
NO TRACK Lt - Out After 12 sec

6 F 04 06 RR TRUNNION AND SHAFT
R1 Varying @ 1/2 cps
R2 Varying @ 1/2 cps
PRO

7 F 16 72 RANGE, RANGE RATE
R1 +18900 To +19800 (+19571)(**195.75**)
R2 -00459 To -00541 (-00495)(**-496**)
RNG/RNG RT MON +109 To +198 mm/-459 To
-541 fps (196/-493) (**195.5/-493.5**)

V34E

8 RADAR TEST - OFF (NO TRACK Lt-On, X-PNTR-
Center)

9 VACATZE RR CDU ZERO (10 sec)

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12 V41N72E
N73 R1-04000
R2-04000
N06 R2 00002
V16N72E (Verify FDAI Needles Up & Right)
V44E (TERM DESIG)

13 V41 N72E
N73 R1-00400
R2-00400
N06 R2 00002
V16 N72E (Verify FDAI Needles)
V44E (TERM DESIG)

14 V41N72E
N73 R1-00000
R2-00000
N06 R2 00002
V16N72E (Verify FDAI Needles)
V44E (TERM DESIG)

15 V41N72E
N73 R1-00000
R2-32300
N06 R2 00002
V16N72E
CB(11) PGNS: RNDZ RDR - Open
AC BUS A: RNDZ RDR - Open
V44E (TERM DESIG)

16 ADVISE CSM RR CHECK COMPLETE
CSM ACTIVATE RR XPNDR

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LIGHTING CHECK

2 min

1 EXTERIOR LTG - TRACK, DOCK, OFF
Verify Operation In Each Position
Mount COAS In FWD Window And Check
3 CB(11) AC BUS B: AOT LAMP - Close
Verify Operation
CB(11) AC BUS B: AOT LAMP - Open

*PREP FOR UNDOCKING

2 min

1 Configure Cameras:
Seq Camera (CE368 Mags (L) (T is Stowed),
5mm Lens, f11, 1/250, 8 fps)
STD Hasselblad CE368 Mags (F) (G is Stowed),
f11, 1/250, Handle, Focus As Required)

CSM Configure For Duplex A & RCY ONLY-A
LM Configure For Basic Comm With VHF B
Backup
AUDIO (Both): VHF B - RCY
VHF B XMTR - VOICE
VHF B RCVR - ON

Mount Phasing Pad
DET - SET (Set for RCS Sep Time)
Overhead Hatch - Locked
OVHD DUMP VALVE - AUTO
PRESS REGS ASD - CABIN
LIGHTING: EXTERIOR - DOCK

4 ATTITUDE CONTROL (3) - PULSE
GUID CONT - AGS
MODE SEL - LDG RADAR
RNG/ALT MON - RNG/RNG RT
RATE/ERR MON (Both) - LDG RDR/CMPTA
ATTITUDE MON (Both) - AGS
SHFT/TRUN - +5°
X-POINTER - TII MULT

5 ENG ARM - OFF
X-TRANSL - E JCTS

On

LM

12

13

14

5

6

RDZ-36

BAL CPL - ON
DEAD BAND - MIN
MODE CONTROL - ATT HOLD
TTCA (BOTH) - JETS
RDZ RDR - LSC

Verify Undocking Attitude As:

SCDG ANGLES: ~~000~~ 120°
 ~~100~~ 195°
 ~~000~~ 000°
FDAI ANGLES: R 180°
 P 016°
 Y 300°

CB(11) R/D INST A - Close

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13

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16

CONFIGURE PGNS & AGS

2 min

1 SET (WAP V48E
 N46 01011E
 N41 LM WT ~~100000~~

V37E 00E
 V76E
 MODE CONTROL - ATT HOLD

3 V37E 30E 1st On 2nd On
 N35 T10 (T00:26:00)(T01:55:00)
 N82 $\Delta V_X(LV)$ (+41590) (+49698)
 $\Delta V_Y(LV)$ (-49602) (-49538)
 $\Delta V_Z(LV)$ (+00406) (+00471)
 N42 NA
 HP
 AV

N45 M, TFI, MGA

410 + SE EXTERNAL AV

450 + E

451 + C

452 + C

247 R → 411 + IE XPS BURN

400 + IE GUIDANCE STEERING

500R

Verify Both ENG STOP ab - Reset

PIPA BIAS CHECK

3 min

1 DET - Zero
 Request CSM Null Rates $< .1^\circ/\text{sec}$
 Request CSM Go CMC MODE - FREE

V25N21E, E, E, E/start DET

VOGE
 06 21 XYZ PIPA PULSES (pulses)

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 Basic Data
 Channel

LM-3

13

14

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16

At T+32 sec - ENTR
 T+32 sec (X)R1 _____ (+XXXAB)
 (Y)R2 _____
 (Z)R3 _____

Notify CSM Test Complete

V06 NOTE, 1452E (R1-Review X BIAS)E (+AB000)
 1454E (Review Y BIAS)E
 1456E (Review Z BIAS)

V21 NOTE
 F 21 01 LOAD 1452E(Calculated X BIAS) E,E (+AB000)
 1454E(Calculated Y BIAS) E,E
 1456E(Calculated Z BIAS) E
 (Use Same Sign As In Measured Bias
 Of Step 4)

PREP FOR TRANSFER

40 min

1 Stow the following in ISA:
 CDR RDZ Procedures
 LMP Rndz Charts & Procedures
 EVA Checklist (S) & G4N Dictionary
 A11 Exposed And Unexposed Film (2/16mm)
 Mags (L) & (T), 2/70mm Mags (F) & (G)
 STD Hasselblad ~~Camera~~
 SEQ Camera & 5mm Lens
 Camr Handle
 Radiation Survey Meter
 Passive Radiation Dosimeter
 Verify Tunnel Pressurized With CSM
 PRESS REGS AAB - EGRESS
 OVHD CABIN DUMP VALVE - OPEN

2 Stow OPS On Floor
 Stow Helmet Stowage Bags On Deck

~~Unstow PLSS, Remove LICW Cartridge~~
~~And Stow In Bag~~

~~Stow PLSS On CDR Side Wall~~

3 Open Hatch
 OVHD CABIN DUMP VALVE - AUTO

Remove Drogue And Probe And Stow On
 LMP Side Wall

CDR TRANSFER TO CSM UMBELICAL

5 min

~~Disconnect CSM Umbilical~~
 CDR SUIT ISOL - SUIT DISC
 CSM Suit Isolation Power Control - To Suit: Flow
 and Aside Suit Power - OFF
 Connect To CSM Comm Line (Address: Stowage)
 Notify CSM Then Conduct Comm Check

2 Disconnect LM Hoses And Stow
 CDR Transfer To CSM with ISA & CDR RDZ
 Checklist

3 Docking Tunnel Index Angle $\delta\theta$ _____

4 Unstow PLSS, Remove LICW
 Cartridge, & Stow In Bag & Pass To CSM
 Stow PLSS On CDR Side Wall

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LM-2

14

15

16

LM SWITCH CLOSEOUT FOR JETTISON

1 min

ORPEAL

LTG - OFF

MASTER ARM - ON

CDR Audio

S-BAND T/R - OFF

RELAY - OFF

GUID CONT - 90NS

THR CONT - AUTO

MAN THROT - CDR

ENG ARM - OFF

ASC He REG 1A2 tb-gray

ABORT And ABORT STAGE - Flush/Guarded

SYS AAB ASC FUEL & ASC OXID (4)-tb-bp

SYS AAB NUD 1,2,3,4 (8) - tb-gray

CRSPD - tb-bp

SYS AAB MAIN SDV - tb-gray

ATTITUDE MON - AGS (LMP)

GLYCOL - PUMP 1

02/H2O NTY MON - C/W RESET

ATTITUDE CONTROL (3) - MODE CONT

MODE CONTROL - ATT HOLD

RCS SYS A/B-2 NUD 1,2,3,4 - AUTO

EXTERIOR LTG - ~~ON~~ OFF

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Changed

LM

7 NCA/J4 JET (2) - ENABLE
TTCA/TRANSL (2) - ENABLE

8 INVERTER - 2

~~BAT 446 NORMAL FEED (2) - ~~ENABLE~~~~

BAT 446 BACKUP FEED (2) - ~~ENABLE~~ ON 44-gray

BAT 446 NORMAL FEED (2) - OFF/RESET tb-bp

9 Audio (LMP)

S-BAND T/R - OFF

RELAY - OFF

10 S-BAND - PH, PRIM, PRIM, OFF, PCM,
RANGE, OFF, HI

TAPE - OFF tb-bp

S-BAND - FWD ~~OFF~~

11 SUIT GAS DIVERTER - PULL/EGRESS

CABIN REPRESS - CLOSE

PLSS FILL - CLOSE

DES O2 - CLOSE

ASC 1 O2 - CLOSE

ASC 2 O2 - OPEN

SUIT ISOL (CDR) - SUIT DISC

SUIT CIRCUIT RELIEF - AUTO

CABIN GAS RETURN - EGRESS

12 CABIN RELIEF AND DUMP (2) - AUTO

13 OFI PRIMARY - ON, SECONDARY - OFF

12 LMP TRANSFER TO CSM UMBILICALS 4 min

1 Connect To CSM Umbilical

LMP SUIT ISOL - SUIT DISC

Request CSM Set LMP Suit Flow Control

To Suit Full Flow And Audio Power - Off

2 Connect To CSM Comm CDR (Audio, Biomed)

Notify CMP Then Conduct Comm Check

VHF - OFF, OFF, OFF, OFF

Disconnect LM Hoses And Stow

PRESS REG ABB - EGRESS

Basic Date Changed 24, 1969

3 Remove Tape Recorder & Pass To CSM
Receive And Stow ISA On Floor

V37E00E

UPDATE AGS

ALIGN AGS

~~100+16~~
~~221A~~ ~~100+16~~ ~~221A~~ ~~100+16~~ ~~221A~~

V37E 42E

ENTR

F 50 18

F 50 19

CSM MNVRS

Verify Attitude

V37E00E

Verify CSM Min Deadband

MODE CONTROL - AUTO

SYS AND ASC FEED 2(2) - OPEN tb(4)-gray

SYS AND MAIN SOV - CLOSE tb-bp

FINAL CABIN CLOSEOUT

INVERTER - OFF

CB(11)

Row 1: All Open

Row 2: All Open Except:

RCS SYS A: QUAD 4,3,2,1 TCA(4) - Close

Row 3: All Open Except:

INST: SIG CONDOR 1 - Close

STAB/CONT: ATCA (9GNS) - Close

AELD - Close

ENG CONT - Close

Row 4: All Open Except:

HEATERS: RCS SYS A/B-1:

QUAD 4,3,2,1(4) - Close

ECS: GLYCOL PUMP 2&1 - Close

GLYCOL PUMP AUTO TRNFR - Close

COMM: UP DATA LINK - Close

PGNS: LGC/OSKY - Close

IMU STBY - Close

IMU OPR - Close

Row 5: All Open Except:

EPS: BAT FEED TIE (2) - Close

XLVNA 905 TIE - Close

ASC ECA - Close

LM-3

DC BUS VOLT - Close
R/D INST A - Close
D/FI PWR - ON
R/D INST B - Close

Row 1: CB(16)
All Open Except:
RCS SYS B:
QUAD 1,2,3,4 TCA (4) - Close
TEMP/PRESS DISP-FLAGS - Close
POGS DISP - Close

Row 2: All Open Except:
LTG: FLOOD - Close
TRACK - Close
MASTER ALARM - Close
STAB/CONT: AEA - Close
ENG ARM - Close
ASA - Close
AELD - Close
ATCA - Close
INST: CMEA - Cycle Open Then Close
SIG SENSOR - Close
PCM/TE - Close
SIG CONDOR 2 - Close

Row 3: All Open Except:
COMM: PRIM 5-BD PWR AMPL - Close
PRIM: 5-BD XMTR/RCVR - Close
PMP - Close

Row 4: All Open Except:
HEATERS: RCS SYS A/B-2
QUAD 1,2,3,4 (4) - Close
EPS: DISP - Close
DC BUS VOLT - Close
ASC ECA - Close
XLUNAR BUS TIE - Close
BAT FEED TIE (2) - Close

Ingress CSM And Secure Hatch

Basic Date Feb. 24, 1969
Changed

FIRE/SMOKE IN CABIN (Not In Suit Loop)

PRESS REG A/B - EGRESS
SUIT GAS DIVERTER - EGRESS
Check POWER/TEMP MON For Excessive
Current, Remove Power From Affected
Bus
Use Fire Extinguisher As Required
Don Helmets And Gloves
SUIT FAN - Redundant Fan
Combustion Products Should Be
Considered Toxic. Smoke And
Contaminants Must Be Removed
From Cabin Before Removing
Helmets By Purging or Dumping
Cabin.

If Fire Persists
Prepare To Dump Cabin (If Ascent
Stage Only, Insufficient O2
Available For Repress)
Visually Perform Suit Integrity Check
CABIN GAS RETURN - EGRESS
CABIN DUMP (Fwd) - Open, Then Auto
At 3.2 psia (Verify Suit Press -
3.6-4.3 psi)
Cabin Dump Valve Open, Until Cabin
Press 0 psi

When Fire Goes Out
SUIT GAS DIVERTER - PULL/EGRESS
SUIT CIRCUIT RELIEF - AUTO
CABIN GAS RETURN - EGRESS
CO2 Canister - MID Position
PRESS REG B - EGRESS
PRESS REG A - DIRECT O2 (until ARS
Clears Suit Press Will Increase To
About 5.8 psia)
CO2 Canister Sel - PRIM

EMER-2

FIRE/SMOKE IN SUIT LOOP (MAY BE IN CABIN ALSO)

SUIT ISOL (Both) - SUIT DISC

SUIT VAN - OFF

PRESS REG ABB - CLOSE

*Combustion Products Should Be Con-

sidered Toxic. Smoke And Contam-

*inants Must Be Removed From Cabin *

*Before Removing Helmets By Purging *

*or Dumping Cabin *

Remove Helmet & Gloves

SUIT CIRCUIT RELIEF - CLOSED

CABIN GAS RETURN - EGRESS

SUIT GAS DIVERter - PULL/EGRESS

Isolate SUIT LOOP Electrically

CB(11)EC5: SUIT FAN 1 - Open

CB(16)EC5: SUIT FLOW CONT - Open

SUIT FAN 2 - Open

SUIT FAN 4P - Open

DIVERter VLV - Open

CO2 SENSOR - Open

CABIN REPRESS - MANUAL As Necessary

To Maintain Cabin Pressure And

Replenish O2

When Fire Goes Out

If Cabin Contaminated - Purge As Necessary

CABIN REPRESS - MANUAL

CABIN DUMP VALVE - AUTO

If Ascent Stage Only, Closely Monitor

O2 Supply

Basic Data
Changed

EMER-3

WARNING LIGHTS LM-3

AGS

1. AEA Test Mode Discrete Signals Fail Condition
2. ASA Heater Fails On Causing A Temp Sensor To Open The +12 vdc Supply
3. Power Supplies Out Of Limits

1. Switch To PGNCs Control

2. Perform AGS Self Test 412R (+1 Test Passed)

ASC PRESS

1. He TANK < 2775 psia (Inhibited after Staging)
2. FUEL, OXID TANK < 120 psia

1. Cross-Check PRPLNT TEMP/PRESS MON - ASC

2. Shut Down APS When Press < 120 psi

Close ASCENT He REGS 1&2 And Check For Leak

CABIN

1. Cabin Pressure < 3.7-4.45 psia (Light Disabled In Egress Position)

1. Cross-Check Cabin Press, Suit Press, & Cuff Gages

2. Close CABIN DUMP VLVs, Check PRESS REGS ABB - CABIN, CABIN REPRESS - AUTO

3. Don Helmet & Gloves, Then Turn CABIN REPRESS - CLOSE And PRESS REGS ABB - EGRESS, CABIN GAS RETURN - EGRESS, SUIT GAS DIVERter - PULL/EGRESS To Troubleshoot

CES AG

1. CES AG Pwr Supplies Out Of Tolerance

QUID CONT - PGNCs

(No Rate Damping Or Attitude Control Or No Manual SPS Throttling In AGS)

Basic Data
Changed

EMER-4

CES DC 1. CES DC Pwr Supplies Out Of Tolerance

1. GUID CONT - PGNS
(No DPS Throttling In PGNS Or AGS, AGS Direct Still Available, DPS May Go To 100%, Deadband Inoperative)

DC BUS 1. Either Or Both DC Buses < 26.5 V

1. Check For BATTERY CAUTION LT, BATT Or DC BUS Component Lights, Check All BAT & BUS Voltages, And Reset All BAT tb's.
2. If No Other Lights - Separate Buses By Opening CROSS TIE BAL LOADS cb And Observing PWR/TEMP MON. If Volts And Amps Not Normal Power Down Low Bus, Open All cb's, Check For Bus Or Feeder Short By Observing DC BUS FAULT LT (If SE BUS) And PWR/TEMP MON (If CDR's BUS)
- 3a. If DC BUS FAULT LT ON (And Unstaged) Indicates ECI Or DFR OPEN - Use BAT 6 NORMAL CDR FEED-ON.
- 3b. If DC BUS FAULT LT ON (And Staged) Power Down Bus, Open All cb's (Except BAT FEED TIE(2) And DC BUS VOLT). If DC BUS FAULT LT Still On Indicates BAT 6 Lost, Reconfigure With BAT 6 Backup Feed. If DC BUS FAULT LT OUT - Indicates Failed ECA.
4. If BATTERY CAUTION LT And DC BUS LT ON - CDR Bus Power Lost - Check BAT 6 - If Only One BAT bp (Unstaged Only) Remove Other BAT And Reset The bp BAT. If Both BATS bp (Or 1 Bat bp Staged) Check For Bus Or Feeder Short By Opening All cb's (Except DC BUS Volts) And Observing DC BUS FAULT LT - If LT OFF, CDR BUS FEED SHORT.
5. (For LMP BUS POWER FAILURE, C/W PWR CAUTION LT WILL BE ON) And Flood And Integral Lights Will Be OFF - See Procedures Under C/W PWR Procedures

Basic Date
Changed

EMER-5

1. Burn Time < 2 Min P 251
2. FUEL Or OXID < 6%
(Only ARMED When DPS's ON)

1. Cross-Check PROP QTY
2. Shutdown DPS Before Depletion

DPS REG 1. He Downstream Of REGS < 200 Or > 260psi

1. Close REG 1 & Open REG 2
2. If Press Decay Continues Below 140 psi Shutdown DPS
3. Close REG 2

ISS 1. IMU Fails, ICDU Fails, Or PIPA Fails During Thrust

1. Switch To AGS Control
2. Perform MANUAL Engine Shutdown

LGC 1. LGC Prime Pwr, Scalar, Or Counter Fails

1. Switch To AGS Guidance
2. Perform MANUAL Engine Shutdown

REG A WEG 1. He REG PRESSURE < 165 Or > 205 psi

1. Cross-Check PRPLNT FUEL MANF And OXID MANF Pressures
2. If Press Below 140 Or Above 205 psi, Turn Off SYS A MAIN SOV And Use CROSSFEED

REG B WEG 1. He REG PRESSURE < 165 Or > 205 psi

1. Cross-Check PRPLNT FUEL MANF And OXID MANF Pressures
2. If Press Below 140 Or Above 205 psi, Turn Off SYS B MAIN SOV And Use CROSSFEED

Basic Date
Changed

RCS TCA

1. One Or More Thrusters Fail Off
2. Colinear Thrusters Fire Simultaneously
3. Detects Thruster Failed On If Rate Feedback In Effect

1. If Only One Red Flag, Close QUAD VALVE
If More Than One Red Flag In Same SYS, Open MAIN SOV or ASC FEED In Affected SYS A or B
2. Complete Translation If Translating
3. If Thruster Firing Continuously, Turn Off QUAD VALVE With Red Flag (This Would Fix Stuck-On Thruster)
4. If Not Continuously firing Recycle All RCS VALVES To Normal Position - Use TCA cb's And Rate Command Attitude Control To Troubleshoot

SUIT/FAN

1. Suit Pressure < 3.12 psi
2. Suit Fan #2 Fails While It Is Selected

Cabin Depressurized

- 1a. Cross-Check With SUIT And CABIN PRESS, And With PGA CUFF GAGE
- 1b. Check PRESS REG ABB - EGRESS. If Press Still Too Low Set PRESS REG ABB - DIRECT O2 Intermittently To Hold Suit Press Up Until Cabin Can Be Repressurized
- 1c. If Suit Isol Valves Closed And Suit Integrity OK, Pull CB(16) ECS: Suit Flow Cont - Open & Suit ISOL VLVS To Suit FLOW

Cabin Repressurized

- 2a. Cross-Check SUIT And CABIN PRESS And CUFF GAGE
- 2b. Check Suit Flow If SUIT FAN 2 Selected
- 2c. If Suit Isol Valves Closed - Pull CB(16) ECS: Suit FLOW Cont - Open & Suit ISOL vlvs To Suit FLOW

Basic Date
ChangedBasic Date
ChangedCAUTION LIGHTS LM-3ASC HI REG

1. He Manifold Press > 220 psi

Continue Engine Burn And Close Both ASC He REG 1 & 2

2. Cross Check ASC PRESS
3. When He Press Drops Below 220, Open REG 1 - If Light Comes On Again Close REG 1 And Open REG 2

ASC QTY

1. When About 10 Sec Of Propellant Burn Time Remains

1. Stop ASC Fuel And ASC Oxid Feed To RCS
2. Cross-Check With ASC He Pressure
3. Shutdown ASC Engine

BATTERY

1. Bat Reverse Current > 10 Amp For 4-6 Sec
2. Bat Temp > 145°F
3. (Bat Overcurrent Trip Will Result In Additional Loss)

1. Check All Bat cb's. If bp, Try Resetting Bat OFF/RESET Then On
2. Check All Bat Positions And Note On Which Bat The Bat Fault Component Lt Illuminates
3. Remove Faulty Bat From Bus

C/W PWR

1. No C/W Warning Pwr
2. Illuminates With Loss Of Entire LMP Bus

- 1a. No Other C/W Lights (C/W Failure) - RESET C/WER cb
- 1b. DC Bus Fault Lt = ON, Flood And Integral Lighting OFF (LMP Bus Failure)
- 2b. LMP AUTO CONT - BU, Pwr Up INV 2, And Open All cb's On LMP Panel (16)
- 2c. If DC Bus Fault Lt Goes Off, Bat To Bus Feeder Open. Set BAT S NORMAL SE FEED-ON, Close All cb(16)

EMER-8

- 2d. If DC Bus Fault Lt Stays On, Take Each Bat Off Line In Sequence And Monitor The DC Bus Fault Lt. When Lt Goes Out It Signifies Faulty Bat Removed. Reconfigure With ASCENT BAT 5 And/OR CROSS TIE BUS cb's.
- 2e. If DC Bus Fault Lt Stays On With Both Bats Removed, Open Both BAT FEED TIE cb's. If DC Bus Fault Lt Goes Out It Indicates A Feeder Short. Reconfigure With Bat 5. If Lt On Indicates LMP Bus Short.

ECS

1. Suit Fan $\Delta P < 6"$ H₂O
2. CO₂ Partial Pressure > 7.6 mm Hg
3. H₂O Sep Speed < 800 rpm
4. Glycol Pump $\Delta P < 3$ psid

1. Cross-Check Component Lights And Select Redundant Component
2. If No Component Lights On, CMEA Failure

ED RELAYS

1. Light Illuminates If Contacts Of Master Arm Relay or Staging Sequence Relays Fail Closed (Both Stage System Lights Should Be On When Master Arm - On, And Off When Master Arm - Off)
1. Do Not Turn On Master Arm Switch
2. Attempt To Reset Stage Relay - If Unable Pull Appropriate ED Logic Pwr

GDA/ROCA

1. Light On If There Is Discrepancy Between Gimbal Drive Signal And Gimbal Response During DPS Burn
1. Monitor RCS Duty Cycle And Attitude Rates. If Increasing, Set ENG GIMBAL - OFF
2. If RCS Duty Cycle Still Excessive, Shutdown The DPS

Basic Date
ChangedBasic Date
Changed

EMER-9

GLYCOL

1. Glycol Temp $> 50^{\circ}\text{F}$
2. Glycol Remaining In Accumulator $< 10\%$

Cross-Check Glycol Temp And Press, CABIN And SUIT TEMPS, And H₂O QTY

- 2a. If Glycol Temp $> 50^{\circ}\text{F}$ Observe Rate Of Glycol Temp Increase. If Temp Steady Suspect Thermal Overload Or INST/CMEA Failure
- 2b. If Increasing, Recycle PRIM EVAP FLOW - OPEN
- 2c. If Glycol Temp Continues To Increase, Close PRIM EVAP FLOW And Set PRIM EVAP FLOW #2 OPEN
- 2d. If Glycol Temp Continues To Increase, Activate Secondary Loop By WATER TANK SEL-SEC, GLYCOL INST(SEC), cb GLYCOL PUMP SEC - Close, SEC EVAP FLOW OPEN Then Shut Down Primary Loop
3. If Glycol Temp $< 50^{\circ}\text{F}$ Suspect Low Glycol Quantity, Monitor Glycol Temp, Press Gages

HEATER

1. RR - Temp $< -54^{\circ}\text{F}$ or $> 148^{\circ}\text{F}$
2. LR Heater (CAW Not Operable In LM-3)
3. RCS - Temp $< 119^{\circ}\text{F}$ or $> 190^{\circ}\text{F}$
4. S-BD-Temp $< -64^{\circ}\text{F}$ or $> 153^{\circ}\text{F}$

- 1a. RR Too Hot - Check For Heat Soak, Open Both RNDZ HTR cb's (If Still Too Hot Open CB AC BUS A; RNDZ RDR, Then CB PGMS; RNDZ RDR When RR Not Needed)
- 1b. If Too Cold - Power Up RNDZ RDR
- 2a. LR Too Hot - Open LDG RDR HTR cb & CB PGMS; LDG RDR When LR Not Needed (Note: When CB PGMS Is Closed, Relay Open Line Between CB HTR; LDG RDR & LR Heaters)
- 2b. LR Too Cold - Power Up LDG RDR
- 3a. RCS Too Hot - QUAD A/B-1 & 2 - Open
- 3b. RCS Too Cold - QUAD A/B-1 (or 2) - Closed, RCS SYSTEMS A/B-2 SW - MANUAL
- 4a. S-BD Too Cold - CB COMM: S-BD ANT - Close
- 4b. S-BD Too Hot - CB COMM: S-BD ANT - Open & CB HTR: S-BD-Open (When S-BD not Needed)

INVERTER

1. AC Voltage < 112 Volts
2. Frequency < 398 cps > 402 cps

1. Check All AC Bus A/B: BUS TIE INV 1&2 cb's (4)
2. If Some cb's Open Determine If INV 1 Feeder Short Or AC Bus A Or Bus B Short By Pulling BUS TIE INV 1 cb's And Closing Bus Tie INV 2 cb Then Monitor cb's
- 3a. If All cb's Remain Closed Cross-Check With Pwr/Temp Mon
- 3b. If VOLT In Green - Determine If INV 2 Or CMEA Failure By Powering Up INV 1
- 3c. If Volts Not In Green - Determine If INV 2 Or INV 2 Feeder Short By Powering Up INV 1 - If Any AC BUS A/B: BUS TIE INV cb's Open; Indicates INV 2 Feeder Short
4. If All cb's Do Not Remain Closed - Check For AC Bus Short Or INV 1 Feeder Short By Opening Both AC BUS A/B BUS TIE INV 1 cb's And Closing Both AC BUS A/B BUS TIE INV 2 cb's.

LOG RDR

If Light Comes On It Is A Failure Of The LR Data Good Circuit Or CMEA

O2 QTY

1. Descent Qty < 55
2. Either Ascent O2 Tank Qty < 80% Prior To Staging
3. Ascent O2 Tank #1 Qty < 10% When Staged

1. Cross-Check O2 Qty Gage and Cabin Press
- 2a. If Cabin Press Normal and O2 Leak Outside Cabin Or Cabin Leak, Close Dump Vlv's And Put Cabin Repress - Close And PRESS REGS A/B - EGRESS
- 2b. If Cabin Press High - Check PRESS REGS A/B, CABIN REPRESS, PLSS VLV'S For Possible Failures
3. If Descent O2 Lost, Go To ASC 1 And Close CABIN REPRESS, SUI7 GAS DIVERTER - PULL/EGRESS And CABIN GAS RETURN - EGRESS

PRIM AMPS

1. Lt On If Either RCS Bias Pwr Supply To The Prim Preamp'd & Jet Drivers Out Of Limit (One From cb ATCA And One From cb ATCA [PGNS])

1. If Both Bias Voltages Out Of Tolerance There May Be Sporadic RCS Fieings

RCS

1. No Tank Press < 1700 psia

1. Cross-Check RCS He PRESS, [PRPLNT, And FUEL MANF And OXID MANF PRES] (The Bracketed Positions Are Useful Only If There Has Been A Propellant Leak Followed By A Bladder Rupture). Cross-Check RCS Qty (Qty Gage Uses He Press)
2. Use Both SYS When FUEL or OXID MANF press 140 psia, Close Bad Systems MAIN SOV For RCS Burns, Use ASC FEED Or CROSSFEED from Good RCS (RCS Fuel And Oxid In Failed System Unusable)

RNDZ RDR

1. When RR Is In Auto Track Mode And Loves CSM Lock-On (NO TRACK LT Should Be On Also)

1. Determine If RR Is Tracking CSM
2. Check For Momentary Data Lost By Selecting MODE CONT - SLEW Then AUTO TRACK
3. Check AGC And XMTA PWR, If Low, Cycle cb PGNS-RNDZ RDR To Eliminate Chrons
4. Attempt Reacquisition With P20 or Visually With AUTO TRACK Or With AGS 400 + 2 And AUTO TRACK Check CSM Transponder

H2O QTY

1. Descent H2O Qty < 14%
2. Either Ascent H2O Tank Qty < 95% Unstaged
3. H2O In Ascent Tank #1 And #2 Differ By > 15%

Basic Data Changed

Basic Data Changed

EMER-12

1. Cross-Check With H2O Gage
2. Verify H2O TANK SELECT in Proper Position
3. Verify SEC EVAP FLOW And PRI EVAP FLOW #2 - CLOSED
4. Monitor Glycol Temp And When > 50° Select ASC H2O Tank

COMPONENT CAUTION LIGHTS

DC Bus Fault

Illuminates When Voltages Between Buses Differs Fully Bright When Voltages Differ By > 18 Volts. When All cb's Are Opened On A Suspect BUS, Then BUS CROSS - TIED To the Other Bus. Light On Will Indicate A Bus That Exists.

Battery Fault

Illuminates When That Specific Battery Temp > 145°F, Reverse Current > 10 Amps For 4-6 Sec, Overcurrent Has Disconnect Bat > 150-200 Amp.

Range/Range Rate Power Failure Light

Loss Of:

1. Range
2. Range Rate
3. PCM/TE
4. AC Power To Meter

Suit Isolation Valves

Closes At 3.11 psia. Cabin Repress Will Be Activated At The Same Time. (If Enabled)

Cabin Repress

Will Activate At 3.70-4.45 And Close At 4.40-5.0.

UPDATE PADS, PHOTO LOGS, CREW COMMENTS

Basic Date
Changed

Basic Date
Changed

LM P27 UPDATE

V		V		V		PURP
INDEX		INDEX		INDEX		
						01 306
						02 307
						03 310
						04 311
						05 312
						06 313
						07 314
						10 315
						11 316
						12 317
						13 320
						14 321
						15 322
						16 323
						17 324
						20 325
						21 326
						22 327
						23 330
						24 331
	*	*	*	*	*	T
		*	*	*	*	4(+N)
	*	*	*	*	*	1(+E)
*	*	*	*	*	*	NAV
						H CHECK

REMARKS

P27
UPDATE

LM P27 UPDATE

V		V		V		PURP
INDEX		INDEX		INDEX		01 306
						02 307
						03 310
						04 311
						05 312
						06 313
						07 314
						10 315
						11 316
						12 317
						13 320
						14 321
						15 322
						16 323
						17 324
						20 325
						21 326
						22 327
						23 330
						24 331
T		T		T		T
+ (+N)		+ (+N)		+ (+N)		+ (+N)
+ (+E)		+ (+E)		+ (+E)		+ (+E)
H NAV CHECK		H NAV CHECK		H NAV CHECK		H NAV CHECK

REMARKS

P27
UPDATE

LM P27 UPDATE

V		V		V		PURP
INDEX		INDEX		INDEX		01 306
						02 307
						03 310
						04 311
						05 312
						06 313
						07 314
						10 315
						11 316
						12 317
						13 320
						14 321
						15 322
						16 323
						17 324
						20 325
						21 326
						22 327
						23 330
						24 331
T		T		T		T
+ (+N)		+ (+N)		+ (+N)		+ (+N)
+ (+E)		+ (+E)		+ (+E)		+ (+E)
H NAV CHECK		H NAV CHECK		H NAV CHECK		H NAV CHECK

REMARKS

P27
UPDATE

FLIGHT PLAN UPDATE

TIME

ITEM

4

8

9

10

11

12

13

14

15

F.P.
UPDATE

FLIGHT PLAN UPDATE

TIME

ITEM

F.P.
UPDATE

PHOTO LOG
16mm (CEX368,EF)

START
SET

MAG

Ft

REMARKS (f,t,LENS,FPS)

TIME

REMARKS

LOG

TIME

REMARKS

TIME

REMARKS

PAUL DIDN'T COME ON (STILL IN BRACKET).
(HAWK 310 DESCRIBE)

ALL PERIMETER CHECK - WIP NOT ON EXIT LAMP
BEC TO BLOCKED CAR (D.H.M.)

PAUL TALKED BY: ALL NOT IN STANDBY ALTHOUGH LAST
THREE ON EVE DAY

ALL ACTIVATION & REL MIT: ALL WARNING LT
CAME ON FROM DEF TO STANDBY & STAYED
ON. ALSO OFFER CAME ON MANY TIMES WITH CAR
BUT ONLY CA - DE EXCEPT FOR BIAS & HYPERBOLIC IN
NOODLES.

DECKED IN ALIGNMENT: OK

ALL PERIM - STAN PER - NOT MUCH - NICE SMOOTH
BOLLS W/ TO 190

REL NOT FIRE - ALL & REL T.D.'S ON BY THIS
TIME.

REL CHECK - WORKING AT ACTIVATION - USED JET/REL
SIVERTER IN LINE TO JET FOR REL PER
DAMPING JET TO T.S.

TIME REMARKS

ENTRY STATE CHECK:

N.C. - PWR SWITCH OVER AS IN

THE BOOK - CW PWR CAUTION LIGHT

INDICATES SWITCHOVER WITH MONITORING

OF OL & FUEL FLOW AND

DEPT IN LIGHTING; LEFT IN WORK POSITION

VAF ACTIVATION: NO DIFFERENCE IN SOUND

WITH ANY ANTENNA COMBINATION

SHORT TEST IN B-2 COULD COME

SECOND TO DEMONSTRATE SILENTLY

OUTER PWR ACTIVATION: NO PROB - SOUNDED LIKE BUT

A LITTLE HIGHER IN EYE AT START.

CEW CHECKOUT - AL: LIGHTS EXACTLY AS IN

CHECKLIST! I owe you a SPAGHETTI

DINNER AT ALMA'S

CB ACTIVATION: SEVERAL TIMES USUALLY OPT > 60°

PRIOR TO FINISHING ACTIVATION I. HAD TO

CLOSE COILS PRIOR TO PWR & OPEN M.C.

TO VALIDATION: HAD IT ONLY IN A AND FORTN2

IT. HUNG UP ON GRAY - ALSO CHARACTERISTIC

GETS TO BE GRAY WHILE HOLDING

SWITCH.

SUIT PWR & HAD THE CHECK - HAD FOR COMP AT THIS

RIGHT TIME TO GO ON (THE HUNG UP WAIT)

& WHEN SWITCHING TO ^{OTHER} PWR IT EXACTLY WAS

ALREADY UP TO SPEED WHILE WHAT DIDN'T COME

ON.

SPP & VAF ACTIVATION: - ON RADY DAY THE LMP HUNG

PT & ACAPIT WERE INOP. VOK OK.

OR OK. FOR OPS HTR. OK DIDN'T LIGHT LIGHTS

ALTHOUGH ON EM DAY IT WORKED 1 OUT OF 5

LOG